

Course Name : Physics II Lab

Course Code : PHY 4242

Experiment no : 01

Experiment Name : Familiarization with Simple
Electrical Circuits

Name : Rafiul Aneeen Islam

Student ID : 210041114

Department : Computer Science and Engineering

Section : 1 (Lab group 1B)

Date of Submission : 24/01/2023

Objective :

The purpose of this experiment is to acquaint the students with the fundamentals of electrical circuits. This experiment will help students to perform future experiments. Through this experiment they will learn how to construct circuits and draw circuit diagram.

The other objective of this experiment is to make students familiar with commonly used measuring equipment.

Theory :

An electric circuit is a closed loop network which allows the flow of current. A circuit can contain many components such as resistors, switches, capacitors etc. However a necessary component is the power source. Current flow occurs because of the voltage that the power source creates. While the

flow occurs, voltage drops across resistors. The amount of voltage drop is related to the current flow, amount of resistance and how the resistors are connected.

Circuit Diagram:

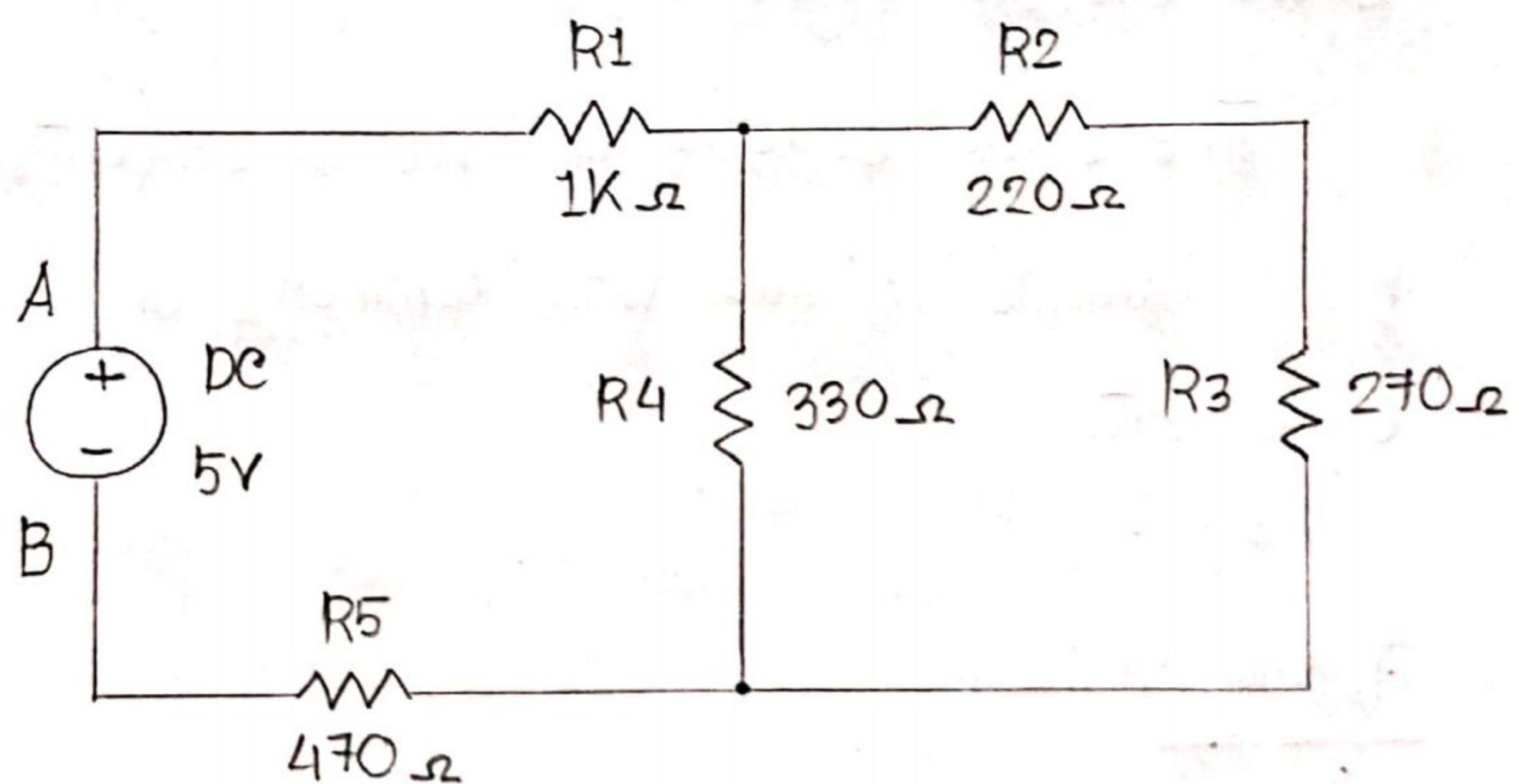


Figure: Circuit diagram for experiment

Data Collection:

Variables taken for resistors R_1, R_2, R_3, R_4, R_5

Variables for voltages across respective resistors,
 V_1, V_2, V_3, V_4, V_5

Variables taken for current flow through each
resistor, I_1, I_2, I_3, I_4, I_5

Name of the circuit element	Value (Ω)	Voltage (V)	Current (mA)
R_1	1000	3.046	3.046
R_2	223	0.270	1.211
R_3	269.7	0.326	1.209
R_4	325.6	0.596	1.831
R_5	460	1.402	3.048

Data Analysis:

The equivalent resistance between A and B is
theoretically,

$$R_{eq} = R_1 + \{(R_2 + R_3)^{-1} + R_4^{-1}\}^{-1} + R_5$$
$$= 1.656 \text{ k}\Omega$$

Practically we get the equivalent resistance,

$$R'_{eq} = 1.661 \text{ k}\Omega$$

$$\text{So, } R_{eq} \approx R'_{eq}$$

Results:

From our data analysis, we can see that the equivalent resistance measured theoretically between point A and B (R_{eq}) and practically (R'_{eq}) are nearly same, as,

$$R_{eq} = 1.656 \text{ k}\Omega$$

$$R'_{eq} = 1.661 \text{ k}\Omega$$

$$\therefore R_{eq} \approx R'_{eq}$$

It can also be observed that voltage drop happens across resistors. The voltage drop for resistors connected in parallel are same. For example, the voltage drop across R_4 (V_4) is the same as the combined voltage drop across R_2 and R_3 ($V_2 + V_3$).

Discussion and Q&A:

The objectives are perfectly achieved by doing this lab. We learned how to draw circuit diagrams and make circuits. We also learned how to build circuits. Through the data analysis and result we learned about equivalent resistance ohm's law. Some things that were noticed during performing the lab -

- i) The number of jumper wires should be reduced.
- ii) The multimeter should be held for some time before the reading becomes stable to get the accurate reading.

Answers to the questions -

- i) The two sides of the DC source A & B are called positive and negative terminal respectively.

The circuit works, which can be seen from the voltage drop across the resistances.

ii) The measurements for the individual resistors are -

$$R_1 = 1000 \Omega$$

$$R_2 = 223 \Omega$$

$$R_3 = 269.7 \Omega$$

$$R_4 = 325.6 \Omega$$

$$R_5 = 460 \Omega$$

The equivalent resistance between A and B

practically is $R'_{eq} = 1.661 k\Omega$

Theoretically, $R_{eq} = 1.656 k\Omega$

$$R_{eq} \approx R'_{eq}$$

So, the practical value is close to the theoretical value. Due to unavoidable reasons there are some discrepancies.

iii) There is some error in the practical value

$$\text{which is, } \frac{1.661 - 1.656}{1.656} \times 100\%$$

$$= 0.302\%$$

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

STUDENT'S NAME: Rafiul Arefeen Islam
STUDENT ID: 210041114

SECTION: 1
GROUP: 1B

COURSE: EEE 4242 (BASIC ELECTRICAL ENGINEERING LAB.)

EXPERIMENT NO. 1

NAME OF THE EXPERIMENT: FAMILIARIZATION WITH SIMPLE ELECTRICAL CIRCUITS.

OBJECTIVE:

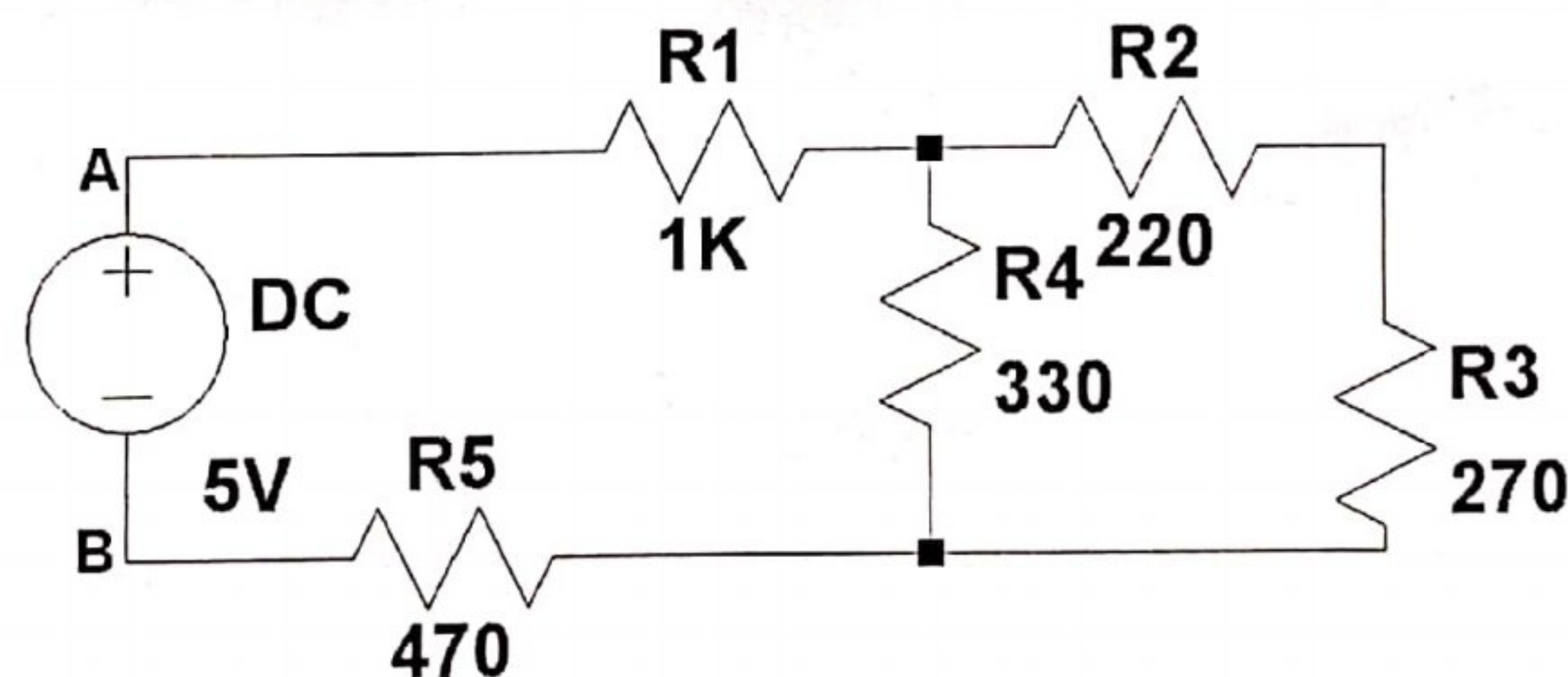
The purpose of this experiment is to acquaint the students with the fundamentals of electrical circuits. This experiment will help students to perform future experiments. Through this experiment they will learn how to construct circuits and draw circuit diagram.

The other objective of this experiment is to make students familiar with commonly used measuring equipment.

LABORATORY TASKS

1. Construct the following circuits and study.

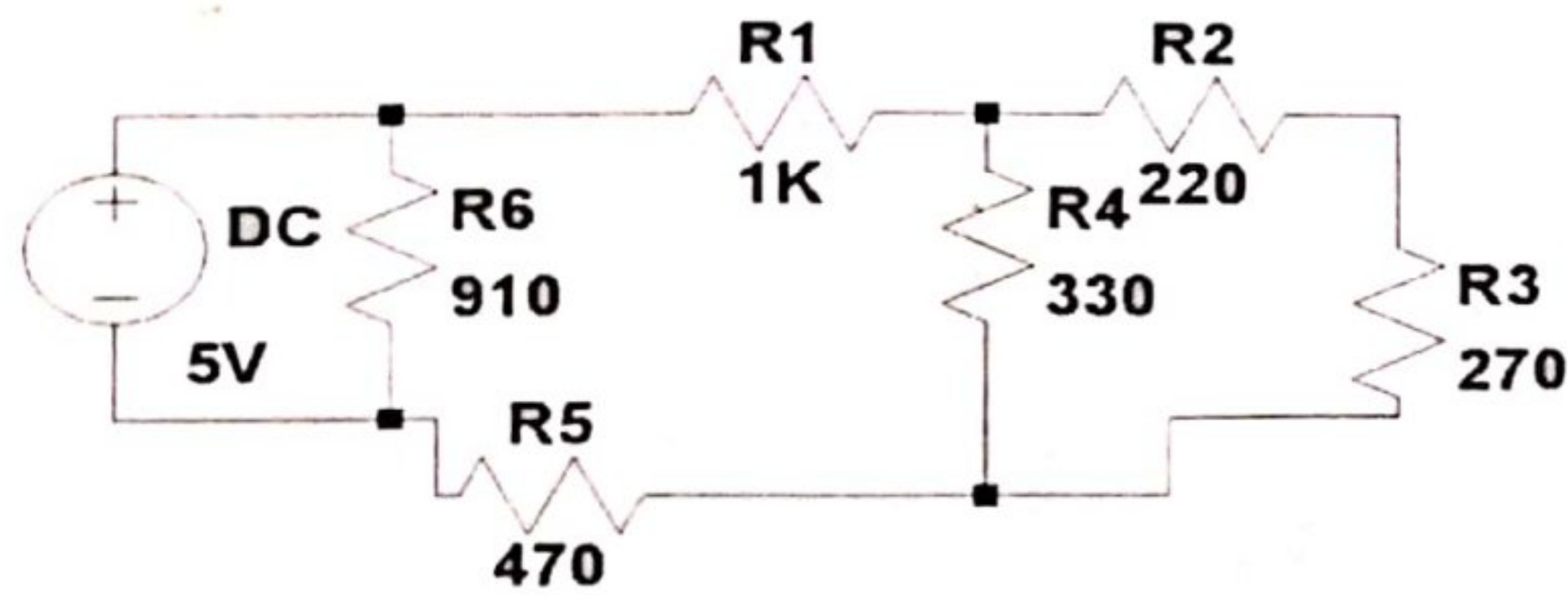
(a)



- Name the two sides of the DC source A & B. Does the circuit work?
- Measure the individual resistance and the equivalent resistance between A and B. Does it agree with the theoretical calculation?
- Comment on your result.

N. B. Before measuring equivalent resistance, disconnect the DC source.

b) Construct the following circuit:



- What is the difference of this circuit with the circuit of (a)?
- Measure current through each branch.
- In which branch current is maximum?

2. Check whether you have taken all necessary data.

3. Before leaving, show completed data sheet to your teacher. You have to include the data with your final report.

Sample Data sheet:

Name of the circuit element	Value	Voltage	Current
R_1	$1k\Omega$	3.046V	3.046 mA
R_2	223Ω	0.270V	1.211 mA
R_3	269.7Ω	0.326V	1.209 mA
R_4	325.6Ω	0.596V	1.831 mA
R_5	460Ω	1.402V	3.048 mA

Report:

- Answer all the questions.
- Show all calculations.
- For each setup identify which elements are connected in series and which are connected in parallel.
- What would be the problem if DC source were not disconnected before measuring the equivalent resistance by multimeter?
- Discuss on your result.